THE FITTER IN YOUR POCKET — DEVELOPER HANDOVER

Takomo Golf × Since AI Hackathon 2026 · compiled Sat 7 Nov 2026, 13:40 Helsinki

This document is the full working context from the ideation session: research, concept, product data, engine rules, interview script, answer bank, swing-video method, voice-CX rules, brand rules, screen specs, the Step 1 prototype code, and open questions. It is written so Claude Code (or a developer) can build from it without the original chat.

How to use with Claude Code: put this file in the repo root (or `docs/`) and reference it from `CLAUDE.md` with a line like `Read docs/HANDOVER.md before starting any task.` The prototype `step1.html` is a standalone page and can be dropped into `public/` or `prototypes/`.

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1. THE CHALLENGE, IN ONE PAGE

Brief (Takomo): buying irons online should not mean choosing blindly. Make expert fitting guidance feel personal, understandable and accessible wherever a golfer is. An intelligent digital fitter should interpret needs, explain trade-offs, and give confidence without pretending every answer is certain.

What exists today: a questionnaire on takomogolf.com (age, gender, height, wrist-to-floor, glove size, skill/score, 7-iron swing speed or carry, ball flight and miss, iron-characteristic preferences) that outputs head family, shaft, length, lie, grip. 600k+ fittings in six months; over 25% of iron orders go through it. Irons only.

Scope decisions made in the session: - Irons only. English only. - Everything is 7-iron based: speed, carry, tempo, calibration. - Units: yards / mph / ft-in by default; metres / km-h / cm behind a toggle. One internal unit. - Score, not handicap: ask "what do you usually shoot on 18".

Judging weights (from the brief): trust & usability 35% · idea 25% · accuracy 20% (28,016-golfer set with known answers + scoring script) · feasibility 20% (must run on takomogolf.com at scale).

Translation: an opaque model that is 3% more accurate loses to a transparent fitter that is equally accurate and feels like a conversation.

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2. CONCEPT

Hybrid: "a fitter who listens, measures what it can, explains everything, and tells you how sure it is."

Two ways in, one engine, one verdict: 1. **Talk to the fitter** (voice or text, ~10 min): chat → measure together → swing on camera → verdict → Q&A/handoff. 2. **Quick quiz** (~3 min): the same nine inputs as a form, each question says why it is asked, with a live "where your fit stands" rail. Result uses the same verdict card with the shaft marked as a guess and a two-minute path into the camera step.

Both feed the same deterministic engine. The fitter is a better way to collect inputs and explain outputs, not a different recommendation. That keeps accuracy provable on the 28k set.

Build rules (non-negotiable)

- The LLM never picks a spec. Engine decides → confidence module says how sure → LLM narrates and asks.
- Every "why" and every answer is built only from fields in `takomo_products.json` or the verdict object. Anything else gets the decline line.
- The fitter observes swings; it never coaches them (also a Takomo brand rule: "clubs don't fix swings, and we don't say they do").
- Accuracy ≥ today's tool on the 28k set — same rules, better inputs.
- Disclose AI in the first line. Human-fitter exit one tap away, with context passed on.
- Order is a payment-class action: read the full spec back, explicit yes, say "ordered" only after Shopify confirms.

Architecture

```
Client (Shopify page / PWA: voice in+out, camera, forms)
   ⇅
Conversation layer (LLM + voice persona) — asks, reads back, narrates
   ↓ writes stated inputs                ↑ reads verdict object
Golfer profile — every input tagged said / measured / seen / sim / estimated, with an error range
   ↑ evidence: swing video (tempo ratio, posture, swing length, speed ±), measurements (read back), sim
numbers (photo/CSV)
   ↓
Fit engine (today's rules + product DB, deterministic)
   ↓
Confidence module — rerun engine across each input's error range; a flip = close call
   ↓
Verdict object — spec · why per line · confidence per line · runner-up · close_call
```

Verdict object (API contract between engine and UI)

```
{
  "head":   {"value":"201-mkii","why":"...","confidence":"high","runner_up":"201t-mkii"},
  "shaft":  {"value":"kbs-tour-stiff","why":"...","confidence":"medium"},
  "length": {"value":0,"why":"...","confidence":"high"},
  "lie":    {"value":1,"why":"...","confidence":"high"},
  "grip":   {"value":"standard","why":"...","confidence":"high"},
  "close_call": {"spec":"head","options":["201-mkii","201t-mkii"],"reason":"inside swing-speed margin"},
  "inputs": {"score":{"v":88,"src":"stated","range":4}, "speed_mph":
{"v":78,"src":"video+carry","range":8}, "...":"..."}
}
```

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3. PRODUCT DATA (FROM `TAKOMO_PRODUCTS.JSON`, AS OF 2026-10-13)

Heads (5). Ratings are Takomo's relative 0–5 scales, not measurements.

HEAD	TARGET SCORE	7I LOFT	7I OFFSET	DIST	FORG	WORK	DISTCTL	FEEL	OFFSET
101 MKII	over 92	29.0°	3.4 mm	5	5	2	2	0	5
201 MKII	85–92	31.0°	2.77 mm	4	4	3	3	0	3
201T MKII	78–84	33.0°	2.46 mm	3	3	4	4	1	2
301 CB	71–77	34.0°	1.8 mm	1	2	5	5	2	0

HEAD	TARGET SCORE	7I LOFT	7I OFFSET	DIST	FORG	WORK	DISTCTL	FEEL	OFFSET
301 MB	under 71	34.0°	1.8 mm	1	1	5	5	2	0

Stock shafts for every head: KBS Tour, KBS Tour Lite. Default: kbs-tour-regular (MKII) / kbs-tour-regular-301 (301).

Shafts (8 families, ~35 flex records). Each record: flex (L/A/R/S/X), weight_g, tempo_profile (slow/mid/fast or any), flight_profile (low/mid/high), heads.

For MKII heads (101/201/201T): KBS Tour (steel 120–133 g, mid|fast, low|mid) · KBS Tour Lite (steel 105–110 g, slow, mid|high) · Mitsubishi MMT (graphite 57–86 g, mid|fast, low|mid) · True Temper Denali Silver (graphite 53–89 g, slow, mid|high).

For 301 heads (CB/MB): KBS Tour 301 build (slow|mid, low|mid) · KBS Tour Lite 301 build (slow|mid, high) · KBS C-Taper (steel 125–140 g, fast, low) · KBS C-Taper Lite (109–125 g, fast, mid|high) · Takomo Obsidian (graphite 72–111 g, any, low|mid) · Fujikura Atmos Red (graphite 72 g, any, high).

Grips: standard / midsize / oversize (Lamkin). **Length:** −1.5″ … +1.5″ in ½″ steps. **Lie:** −3° (flat) … +3° (upright), 0 standard.

Today's rules (`current_tool_rules`, "a starting point, not a spec"): - Head ← score bucket: <71 MB · 71–77 CB · 78–84 201T · 85–92 201 · >92 101. - Flex ← 7i swing speed (or carry): <60 mph / <120 yd → L (women) or A (men) · 60–70 / 120–135 → A · 71–80 / 135–155 → R · 81–90 / 155–175 → S · >90 / >175 → X. - Graphite if: flex is L or A, or golfer is female, or age ≥ 70. - Tempo answers: smooth → slow, moderate → mid, quick → fast. Flight: low / mid / high. - Length by height: <155.5 cm −1.5″ · <160.5 −1.0″ · <168.5 −0.5″ · <183.5 0 · <191.5 +0.5″ · <196.5 +1.0″ · else +1.5″. Lady flex adds +1″. - Lie: five worked examples only (152/86 → +1; 173/86 → 0; 188/94 → +2; 196/100 → +3; 193/76 → −2). **No formula — ask Takomo.** - Grip by glove: XS/S/M standard · M/L, L midsize · XL oversize. - Stock set (no upcharge) = steel stock shaft, standard length, lie, grip.

Gaps to confirm: lie formula; whether forged heads can be bent after purchase; no price per shaft; no handedness field; the graphite rule is blunt; 301 Combo not modelled.

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4. ENGINE — THE SIX IRON DECISIONS

DECISION	DRIVEN BY	WHERE IT GETS SHAKY → CLOSE CALL	ERROR RANGE TO RERUN WITH
Head	score bucket; miss and goal as tie-break	score within 2 of 77/78, 84/85, 92/93; aspirational score; "players look" with inconsistent strike	±4 self-reported · ±2 from app/handicap
Flex	7i speed, else carry ÷ ~1.9	within 3 mph of 70/80/90; carry overstated; carry vs total; metres as yards	±3 sim · ±6 stated · ±10 carry or driver×0.8 · video narrows to ±8 and cross-checks
Shaft model	steel/graphite rule, tempo, desired flight, head family	self-described tempo ("medium"); desired vs current flight mixed; slicer wants "height"	tempo adjacent bucket if stated, exact if video · flight adjacent
Length	height	within 1 cm of a threshold; arm length ignored	±1 cm stated · ±2 "about"

DECISION	DRIVEN BY	WHERE IT GETS SHAKY → CLOSE CALL	ERROR RANGE TO RERUN WITH
Lie	height + wrist-to-floor; video posture as support	weakest rule; skipped wrist-to-floor → Low	wtf ±1″ measured · ±2″ estimated; default Medium until formula exists
Grip	glove size	M vs M/L; no glove	adjacent size if unsure

Confidence = rerun the rules across each input's error range; if the answer flips, it is a close call and the fitter says so.

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5. INTERVIEW SCRIPT (VOICE MODE; QUIZ ASKS THE SAME NINE AS A FORM)

Every captured value carries a `_source` (sim · measured · video · stated · estimated) which sets its error range. Ask a follow-up only when the answer sits inside a range that would flip a decision. Agent turns ≤ 2 sentences then a question. Every number is read back as text.

#	FITTER ASKS	FOLLOW-UP / ASPIRATION PROBE	EXTRACT → FIELD	DRIVES
0	"Tell me about your game — how long you've played, how often, and what you'd like these irons to fix."	Reflect back: "So: … did I get that right?"	years, rounds/month, score if volunteered, miss, goal → `goal`, `miss`	whys, tie-breaks
1	"What do you usually shoot on 18 — a normal day, not your best?"	Vague → "Break 90 most days?" · "Now, or what you're aiming for?"	number / range / hcp (+72) → `score`, `score_source`	Head
2	"Know your 7-iron swing speed from a sim?" — else "How far does a normal 7-iron carry, in the air?"	"Good one or normal one?" · units · driver-only → ×0.8, tagged estimated	mph / carry yd → `speed_mph` or `carry_yd`, `speed_source`	Flex
3	Not asked in voice — measured from the 7-iron swing video. Quiz: "Smooth and rhythmic, or quick and aggressive?"	Quiz, if "medium": "Would a friend say you rush it?"	slow/mid/fast → `tempo`, `tempo_source`	Shaft model
4	"Irons fly high and land soft, or flatter? Want more height, less, or same?"	Slicer wanting height → "More launch won't fix the slice; offset and lie do more."	`flight_current`, `flight_desired` (rule uses desired)	Shaft model
5	"What's your bad shot — slice, hook, thin, fat, toe, heel?"	"Every round or now and then?"	`miss`, `miss_freq`	Head tie-break, whys
6	"What are you playing now, and what flex?"	"How old? Steel or graphite?"	`current_set`	sanity check, whys
7	Measure: "How tall?" · "Tape from wrist crease to floor, arms relaxed" · "Glove size?"	Read back every number. No tape → skip, lie → Low	`height_in`, `wtf_in`, `glove` + `_source`	Length · Lie · Grip
8	"Right- or left-handed?" · "Lighter graphite or steel?"	Age/gender only if the graphite rule needs them, via the shaft-weight question	`hand`, `graphite_pref`, `age`, `gender`	steel/graphite

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6. ANSWER BANK (SHIP AS GROUNDED Q→A PAIRS; RUN AS REGRESSION TESTS)

One sentence of why, then stop. Long explanations raise trust without raising accuracy.

TOPIC	GOLFER ASKS	BUILT FROM	EXAMPLE LINE
Shafts	Why stiff not regular?	speed bucket + edge distance, tempo, confidence.flex	"78 is right on the line; your quick tempo tips it to stiff. Under 75 on a sim and I switch you back."
	Steel or graphite?	graphite_if, shaft.weight_g, material	"Graphite is ~40 g lighter — easier on the hands, a touch more speed. At your speed steel is the stock choice."
	KBS Tour vs Tour Lite?	weight_g, tempo_profile, flight_profile	"Tour Lite is 15–25 g lighter, suits a smoother tempo, launches higher. Tour is mid-to-quick tempo, flatter flight."
	Lighter shaft = more distance?	swing_length, tempo, weight — no yardage claims	"A little speed, usually. I can't promise yards — nobody honest can without a launch monitor."
Lie & length	What does +1° upright do?	lie_sign_convention, hand, miss	"Starts the ball a touch left for a right-hander and squares the sole for your hand position. Too much and you'd pull it."
	Why not longer, I'm tall?	length_by_height, wtf, lie_examples	"Length comes from height, lie from where your hands hang. Tall but long-armed → standard length, a touch upright."
	Can I change lie later?	Takomo policy (confirm)	"Forged heads bend a degree or two at any fitter. [confirm]"
Grip	Why midsize?	grip_by_glove_size	"Large glove → midsize. Too small and your hands get overactive; too big and it's hard to release. Grips are cheap to swap."
Models	101 vs 201 / 201 vs 201T / CB vs MB	ratings deltas, loft, offset, target_score	"201T gives up one point of forgiveness for one of workability, 2° weaker loft, less offset. Strike it clean most days? Fair trade."
	Am I good enough for 201T/301?	score bucket, miss_freq, video consistency	"Not about good enough — whether you'd enjoy the trade. At 88 with a regular slice, the 201 gives you better rounds."
	Mix CB long irons / MB short?	301 Combo exists on site; not in data	"Yes, that's the 301 Combo. I can't fit the split today — a human fitter can."
Trust	How sure are you?	confidence per spec, close_call	"Sure on length, lie and grip. Head is solid. Flex is the one I'd want a sim number for."
	What if it's wrong? Try first?	30-day returns, 2-year warranty, 60+ demo locations	"30 days to send them back, and demo bays where you can hit both first."

TOPIC	GOLFER ASKS	BUILT FROM	EXAMPLE LINE
	Do you just recommend what sells?	engine = rules; current_set comparison	"Same rules as the quiz — I just explain them. If your current set already matches, I'll say so."
	How is this different from the quiz?	—	"Same engine. I get better inputs — I watched you swing — and I tell you why and how sure I am."
Declines	Swing lessons · other brands · wedges/woods/putters · prices in other currencies · medical	refusal list; always offer the human exit	"That's outside what I can fit today. I can get you 15 minutes with a human fitter — they'll see everything we've done."

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7. SWING VIDEO — TEMPO, POSTURE, CLUBHEAD SPEED

Capture: three swings **with a 7-iron**, **face-on**, phone at hip height, **slow-mo (120/240 fps)**. Live capture adds a framing coach (dashed body outline, orange when ok); upload runs the `quality` gate first. The fitter observes, never coaches. After each swing it says one specific thing it saw — that is the trust moment. Face-on, not down-the-line: hand speed is foreshortened DTL; tempo and hands-high/low work from either.

Pipeline (approach informed by the open-source swing-analyzer PWA — reimplement, do not copy; it has no licence): 1. MediaPipe Pose Landmarker in a Web Worker (GPU, CPU fallback), frames downscaled to ~480 px, sampled at ~60 real-time Hz, capped ~900 samples. Detect fps by playing at 0.2× and reading `requestVideoFrameCallback` timestamps; `timeScale` for baked-in slow-mo. 2. Hand = midpoint of wrists (landmarks 15/16). Fill gaps, smooth (~50 ms). Hand speed normalised by torso length. 3. Impact region = peak hand speed. Top = highest hand point before it. Impact = lowest hand point between top and ~120 ms after the peak. Address = last still run (speed < 8% of peak for ~100 ms) before the takeaway. Finish = first still run after impact with hands raised. 4. Confidence 0–1: penalise if hands rose < 0.8 torso-lengths, downswing not shorter than backswing, or < 70% frames with a pose. 5. Tempo = backswing ÷ downswing (real seconds). Three swings → median; discard swings with confidence < 0.6.

Outputs (schema):

OUTPUT	RELIABILITY	USED FOR	FITTER SAYS
`tempo_ratio`, `tempo_bucket` slow/mid/fast	High	replaces stated tempo → shaft model	"Quick transition — about 2.4 to 1."
`address_hands` high/neutral/low	High	supports lie	"Hands hang low — that's why I'm going a degree upright."
`swing_length` short/full/past-parallel	Good	speed proxy; lighter-shaft reasoning	"Compact swing — a lighter shaft gives a bit more speed."
`speed_est_mph` + `speed_range`	Rough · needs slow-mo	never replaces stated speed; narrows band; cross-check if off by a bucket	"You said 85 — this looks closer to 75. Full-effort swings?"
`consistency`	Rough	forgiveness argument	"Those three were very alike."

OUTPUT	RELIABILITY	USED FOR	FITTER SAYS
`quality` ok / no-hands / too-far / too-dark	High	gate	"Can't see your hands — step back a bit?"
Not from a phone	—	ball speed, launch, spin, strike, face angle	"That's why 201 vs 201T stays a close call until a demo bay or sim numbers."

Tempo buckets (starting point, calibrate): ratio ≥ 3.2 → slow/smooth · 2.6–3.2 → mid · < 2.6 → fast/quick; backswing > 1.0 s → slow regardless. Reject if no address stillness, top-to-impact < 5 frames, or wrist confidence < 0.6 mid-swing. Reference: tour ≈ 3:1 (backswing ~0.75–0.9 s, downswing ~0.25–0.3 s); downswing time alone barely separates golfers — the ratio does.

Clubhead speed (rough, ±8 once calibrated), fused from three estimators: - A. carry → speed via the rule table (~1.9 yd/mph for a 7-iron). - B. peak hand speed from pose × calibrated ratio. Pixel scale from the golfer's known height (shoulder→ankle at address). Requires slow-mo: hands peak ~18–22 mph (TPI tour average 22.0 peak mid-hands); at 240 fps that is ~4 cm/frame; at 30 fps it is 30 cm/frame with blur — unusable, fall back to A. Pros ≈ 5× hands, amateurs with a 7-iron nearer 4× — **fit `club_speed = a × peak_hand_speed + b` at the TrackMan session** on 30–50 paired 7-iron swings; the residual is the honest ±. - C. downswing time + swing length as a sanity prior. - Report as a range, never a point. If A and B disagree by a flex bucket, the cross-check question fires. Do not attempt clubhead tracking this weekend.

What the golfer gets back (so the step feels useful): one specific observation after each swing within ~2 s; their swing with a 3-dot overlay and tempo readout; the cross-check said out loud; a "your swing" card (tempo ratio, posture note, speed range) in the summary — a mirror, not advice.

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8. VOICE CX RULES (FROM RESEARCH)

1. **Heard right.** Confirm every number and name — read back, chip to correct. "Repeating myself" is the #1 frustration (55% in a 455-person 2026 survey); misheard 45%; cut off 47.5%. Test with accents and outdoor noise.
2. **Timing.** No fixed silence timeout; semantic end-of-turn. Barge-in: mute, cancel pending generation, keep the partial in history. Agent turns ≤ 2 sentences then a question. No fake "working on it".
3. **Calibrated.** Speak at the engine's confidence: brisk on High, slower and flagged on Medium/Low. One-sentence whys — longer explanations raise user confidence without raising accuracy (UC Irvine / Nature Machine Intelligence calibration-gap study). Express uncertainty per spec, not inside the reasoning chain.
4. **Honest.** Disclose AI in the first line (80% want it; ~20% fewer hang-ups across 450k calls). Human exit one tap away, with context passed on. Order = payment-class.
5. **On screen.** Every number and spec also as text. Hard-to-hear details never voice-only.

Demo metrics: corrections per session, verified completion, close-call rate — not WER.

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9. BRAND RULES (TAKOMO BRAND BOOK, FEB 2025)

- **Colours:** Midnight Black `#0B0B0B` (never pure #000), Washed White `#F8F8F8`, Tiger Orange `#FF6E01`. Ratio 60/30/10; push black and dark shades; orange is an accent only. Greys used in the prototype: `#161616` surface, `#2A2A2A` line, `#3A3A3A` strong line, `#C8C8C8`, `#8E8E8E`, `#5A5A5A`.
- **Type:** Barlow Condensed for headlines (bold, uppercase, tight); Barlow for body and anything small. Both on Google Fonts.
- **Logo:** T-mark from a swing pattern; letter-spaced wordmark TAKOMO. Use the real SVG from brand assets (the prototype uses the wordmark only).
- **Imagery:** rough, deep/moody, natural; product on black, premium. No saturated fairways / blue skies.
- **Voice:** "the best random ever" — funny, good at golf, doesn't give swing tips, knows a ton and uses it helpfully, candid, hypes good shots, pokes fun at bad ones. Archetype Sage + Jester. Pillars: everyman over authority · funny over plain, boring or cheesy · candid over formal · technical over reductive. "Takomo fans don't want to talk to ChatGPT. They want to talk to a friend." Claim: *Honestly better golf.*
- **Direct constraint:** "We know clubs don't fix a bad swing, and we don't tell our buyers that they will."
- Option to ask Takomo: use the illustrated Forge blacksmith character as the fitter's avatar (sidesteps "pretending to be human").

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10. SCREEN SPECS (MOBILE-FIRST WEBSITE)

Shared spine on every fitter screen: (1) wordmark + units toggle / step label; (2) progress spine chat · measure · swing · your fit; (3) **caption strip** — orange left rule, "FITTER · SPEAKING", live text of what he is saying; (4) **stage** — what he is talking about; (5) mic / type / skip in the same place.

1. **Hello.** He is already talking on load (text types out; audio after first tap). Three doors: Talk to the fitter (orange, primary) · Same thing, typing · Quick quiz. Fine print: no email to see your fit · saved so you can come back · same engine either way. Returning golfer: greeting changes, primary becomes Continue.
2. **The chat.** Caption: "Slice and weak long irons — been there. What do you usually shoot? Normal day, not the one you tell your mates about." Stage: profile grid filling in live (Usually shoot · 7-iron carry · Bad shot · Wants), the field being asked in orange, each tile tagged said/heard, tap to fix; "You said" shows the last transcript.
3. **Measure together.** Caption: "Tape at the crease of your wrist, arms hanging like you've given up on the round. What does it say at the floor?" Stage: height / wrist→floor / glove tiles, figure illustration, big number + "I heard 'thirty-three'" + That's right / Say again. No tape → skip, lie goes to Low.
4. **Swing — live or a clip.** Caption reacts after each swing: "Quick transition — about 2.4 to 1 — and your hands hang low. Noted. One more and I'll stop staring." Stage: camera with dashed frame, REC · swing 2 of 3, tempo readout 2.4 : 1, bottom line "7-iron, facing me. Framing's good — swing when you're ready." Card: what I can see / what I can't. Buttons: Use a clip I have · Skip (he guesses).

5. **Your fit.** Caption reads the line he is on; that row lights up. IRON 201 MKII, runner-up 201T. Rows Head / Shaft / Length / Lie / Grip, each with one why and a pill: Sure (grey) · Hmm (orange outline) · Guess (orange fill). One close call box. Buttons: Ask me anything · Review & order.
6. **Ask, then go.** Q&A chips (Why stiff? · +1° lie does what? · 201 vs 201T · Steel or graphite? · "Fix my slice? → nope, clubs don't"). Fit summary saved, 201 vs 201T open. Exits: Hit both at a demo bay · 15 min with a human fitter (they get the summary first).

Quiz: same caption strip relabelled "why I'm asking"; "Where your fit stands" rail (Head · set, Shaft · open, Lie · set); result = the same card with Shaft as Guess and "Turn the guess into a call · 2 min" into the camera step.

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11. STEP 1 UX SPEC (LOCKED FOR PROTOTYPING)

- Lives on the existing "Find my fit" page. No account, no email, no mic wall before the first line.
- 0 s: page loads, caption strip visible and pulsing. 0.5 s: fitter starts. Text first, voice second — caption types out; audio only after a tap (browsers require a gesture). "Tap to hear him" on the strip.
- Line: "Hey. I'm Takomo's AI fitter — not a person, but I know these irons better than most people do. Ten minutes, I explain every spec, and I tell you when I'm guessing. Talk or type?"
- Doors tappable while he talks; tapping stops him mid-sentence (barge-in).
- Talk → mic permission sheet with context ("He needs to hear you. Nothing is recorded or kept."). Denied → typing, no second ask. Typing = identical fitter, keyboard. Quiz = tertiary.
- Units toggle top right, remembered. Returning golfer: "You're back. Last time we left 201 vs 201T open — got sim numbers, or pick up where we were?" with Continue.
- Must not: pretend to be human; cookie-banner-style wall; autoplay video with sound; onboarding carousel.
- Dev: caption strip is a reusable component; mic requested on tap, never on load; this fixed line can be pre-rendered TTS (zero latency).
- Open calls for the team: speak on load vs after tap; two doors vs three; the fitter's name (not a human one).

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12. PROTOTYPE — `STEP1.HTML`

A standalone mobile-first page: types the caption, speaks with the browser voice (word highlight in sync), barge-in, permission sheet, units toggle, returning-golfer state, demo bar (Replay · Returning golfer · Mute voice). The microphone is simulated (the hosted prototype cannot be granted mic access). Replace `speakLine()` with an ElevenLabs stream and keep the word-highlight logic (ElevenLabs returns word timestamps). Full source in Appendix A.

ElevenLabs key setup agreed: scopes Text-to-Speech (+ Speech-to-Text if used, + Voices read), credit cap, no IP allowlist for the hackathon; key server-side only (or single-use tokens for client streaming); low-latency model for conversational turns.

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13. RESEARCH SUMMARY (MARKET)

Landscape splits into: (A) OEM static questionnaires — PING WebFit (trained on real fittings, "practice round" for in-person), Callaway and Titleist selectors (Titleist adds free virtual consults; its ball tool is praised for explaining *why* and showing the runner-up), TaylorMade Club Recommender; (B) AI co-pilots inside the studio — Club Champion AI Fitter Co-Pilot (~80% accurate by its founder's estimate, i.e. one in five would buy the wrong clubs without live testing), Wilson FitAi (needs a launch monitor), Mizuno Swing DNA (shaft-load sensor), GOLFTEC TECfit; (C) sensor/phone capture — Groove Golf (Apple Watch, "Groove Number"), Sportsbox AI (3D from one phone video; fitting partnerships with LA Golf and Cool Clubs), Arccos (on-course data into fittings); (D) new LLM-style fitters — FitMyGolfClubs (grades your current bag on 8 factors, traceable per-factor reasoning, states its own limits, revenue from analysis not equipment), Eldrick (API/MCP engine from launch-monitor data), True Path Golf, StickFitter, Kinetixx FIT; (E) human-in-the-loop remote — 2nd Swing (video session, Fit Pack). Nobody explains trade-offs well, nobody quantifies confidence, OEM tools are accused of recommending the golfer's aspiration rather than reality. Takomo's small catalogue (5 heads, 8 shaft families) makes explanation tractable; 60+ demo locations are the hand-off for close calls.

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14. PLAN, RISKS, OPEN QUESTIONS

Timeline. Saturday: engine wrapper + verdict object + perturbation confidence; TrackMan session (film every 7-iron swing in slow-mo beside the monitor; collect speed pairs; ask the fitters the questions below); capture module (tempo first, speed second); conversational UI; end-to-end on one demo golfer by evening. Sunday: polish, run all 16 demo golfers, record a walkthrough video as backup, one-page write-up (idea / what the PoC proves / accuracy validation / trust design / next steps), README, hand in.

Roles. Engine & eval · Capture (video) · Conversation & UX (ElevenLabs + caption strip) · Story (write-up, demo script, judging map).

Risks. Pose estimation eating Saturday → video is additive; demo must work without it. LLM saying something a fitter wouldn't → it only narrates the engine; answer bank as regression tests. Chatbot asking too much → ask only when it would flip a decision.

Questions for Takomo (Saturday). How was the 28k set produced (real fittings or synthetic)? Which input causes the most returns/support tickets when wrong? How is lie decided beyond the five examples? Can forged heads be bent after purchase? What does a Takomo fitter say when two heads are a close call? Can the Forge character be the fitter's avatar? Any constraints on LLM use in production (cost at ~100k fittings/month, EU data)?

SOURCES

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Swing video: github.com/JamesBKav/swing-analyzer · github.com/ryanboscobanze/GolfPosePro · developers.google.com/edge/mediapipe/solutions/vision/pose_landmarker/web_js · mytpi.com (measuring swing timing from video; linear kinematic sequence) · dewizgolf.com (five tempo measurements; max hand speed) · deepswing.io/golf-swing-speed-app · arxiv 1903.06528 (GolfDB/SwingNet)

ElevenLabs: elevenlabs.io/docs/api-reference/authentication

APPENDIX A — `STEP1.HTML` (STEP 1 PROTOTYPE, FULL SOURCE)

Standalone mobile-first page. Open directly in a browser; no build step. Browser `speechSynthesis` stands in for ElevenLabs.

```
<title>Takomo Fitter Prototype</title>
<meta name="claude:viewport-fit-sides" content="cover">
<link rel="stylesheet" href="https://fonts.googleapis.com/css2?
family=Barlow:wght@400;500;600;700&family=Barlow+Condensed:wght@600;700;800&display=swap">
<style>
/* Layout: a mobile-first website. One column, full-bleed on a phone; on wider screens the same column
sits centred (max 480px) on the brand ground. Single dark look by choice — it is the brand. */
:root{
  --ground:#1C1C1C;
  --bg:#0B0B0B;
  --surface:#161616;
  --line:#2A2A2A;
  --line-strong:#3A3A3A;
  --fg:#F8F8F8;
  --fg-2:#C8C8C8;
  --fg-3:#8E8E8E;
  --fg-4:#5A5A5A;
  --accent:#FF6E01;
  --accent-ink:#0B0B0B;
  --accent-soft:rgba(255,110,1,.14);
  --display:'Barlow Condensed','Arial Narrow',Impact,sans-serif;
  --body:Barlow,'Helvetica Neue',Arial,sans-serif;
  color-scheme:dark;
}
html,body{height:100%}
body{margin:0;background:var(--bg);color:var(--fg);font-family:var(--body);-webkit-font-
smoothing:antialiased;padding:0;box-sizing:border-box}
.stage{min-height:100%;display:flex;flex-direction:column;align-items:stretch}
.phone{position:relative;width:100%;flex:1 1 auto;min-height:100%;background:var(--bg);display:flex;flex-
direction:column;box-sizing:border-box}
.screen{flex:1 1 auto;min-height:100%;display:flex;flex-direction:column;gap:14px;box-sizing:border-
box;padding-block:28px 24px;padding-left:max(20px,env(safe-area-inset-left,0px));padding-
right:max(20px,env(safe-area-inset-right,0px));position:relative}
.screen[hidden]{display:none}
.demo{position:fixed;left:0;right:0;bottom:0;z-index:5;border-radius:0;padding-left:max(16px,env(safe-
area-inset-left,0px));padding-right:max(16px,env(safe-area-inset-right,0px));padding-bottom:calc(8px +
env(safe-area-inset-bottom,0px))}
```

```
.screen{padding-bottom:88px}
/* wider screens: the same column, centred on the brand ground, like the site */
@media (min-width:700px){
  body{background:var(--ground)}
  .stage{align-items:center;padding-block:32px;padding-left:max(16px,env(safe-area-inset-
left,0px));padding-right:max(16px,env(safe-area-inset-right,0px))}
  .phone{max-width:480px;min-height:calc(100vh - 64px);border-left:1px solid var(--line);border-right:1px
solid var(--line)}
  .screen{padding-left:28px;padding-right:28px;padding-block:36px 24px;padding-bottom:88px}
  .demo{left:auto;right:16px;bottom:16px;border-radius:8px;max-width:420px}
}
.sheet,.scrim{position:fixed}
@media (min-width:700px)
{.sheet{left:50%;right:auto;width:480px;transform:translate(-50%,105%)}.sheet.show{transform:translate(-50
%,0)}}

.top{display:flex;justify-content:space-between;align-items:center}
.wordmark{font-family:var(--display);font-weight:700;letter-spacing:.3em;font-size:15px}
.units{display:flex;gap:2px;background:var(--surface);border-radius:3px;padding:2px}
.units button{font:600 11px/1 var(--body);letter-spacing:.06em;text-transform:uppercase;padding:6px
9px;border:0;border-radius:2px;background:transparent;color:var(--fg-
3);cursor:pointer;transition:background .25s,color .25s}
.units button[aria-pressed="true"]{background:var(--line);color:var(--fg)}
.rule{height:1px;background:var(--line)}

/* The caption strip: the fitter is always saying something, and it is always readable. */
.cap{position:relative;border-left:3px solid var(--accent);padding-left:14px;display:flex;flex-
direction:column;gap:6px;min-height:118px}
.cap-label{display:flex;align-items:center;gap:8px;font-size:11px;letter-spacing:.12em;text-
transform:uppercase;color:var(--accent);font-weight:700}
.dot{width:7px;height:7px;border-radius:50%;background:var(--accent);animation:pulse 1.4s ease-in-out
infinite}
.cap.idle .dot{animation:pulse 1.4s ease-in-out infinite}
.cap.done .dot{animation:none;opacity:.5}
@keyframes pulse{0%,100%{transform:scale(1);opacity:1}50%{transform:scale(.55);opacity:.45}}
.cap-text{font-size:18px;line-height:25px;font-weight:500;min-height:75px}
.cap-text .w{transition:color .12s}
.cap-text .w.on{color:var(--accent)}
.cap-text .w.todo{color:var(--fg-4)}
.hear{align-self:flex-start;display:inline-flex;align-items:center;gap:6px;margin-top:2px;padding:6px
10px;border:1px solid var(--line-strong);border-radius:3px;background:transparent;color:var(--fg-
2);font:600 12px/1 var(--body);letter-spacing:.04em;cursor:pointer}
.hear[hidden]{display:none}
.hear svg{width:14px;height:14px}

.doors{flex:1 1 auto;display:flex;flex-direction:column;gap:10px;justify-content:flex-end}
h1{margin:0 0 6px;font-family:var(--display);font-size:52px;line-height:48px;font-weight:800;text-
transform:uppercase;letter-spacing:-.01em;text-wrap:balance}
.door{display:flex;align-items:center;justify-content:space-between;gap:12px;padding:16px 18px;border-
radius:4px;text-decoration:none;color:var(--fg);background:var(--surface);border:1px solid var(--
line);cursor:pointer;text-align:left;font-family:var(--body);transition:transform .15s,border-color .2s}
.door:active{transform:scale(.985)}
.door:focus-visible,.hear:focus-visible,.units button:focus-visible,.sheet button:focus-
visible{outline:2px solid var(--accent);outline-offset:2px}
.door .t{font-family:var(--display);font-size:22px;line-height:22px;font-weight:700;text-
transform:uppercase}
.door .s{font-size:13px;color:var(--fg-3);margin-top:3px}
.door.primary{background:var(--accent);color:var(--accent-ink);border-color:var(--accent);padding:18px}
.door.primary .t{font-size:24px;line-height:24px}
.door.primary .s{color:var(--accent-ink);opacity:.85;font-weight:500}
.door svg{flex:none;width:26px;height:26px}
.door .arrow{font-size:20px;color:var(--fg-3)}
.fine{font-size:12px;line-height:16px;color:var(--fg-4);text-align:center;margin-top:4px}

/* entrance: from a visible resting state, short */
.rise{animation:rise .5s cubic-bezier(.2,.7,.2,1) both}
.rise.d1{animation-delay:.08s}.rise.d2{animation-delay:.16s}.rise.d3{animation-
delay:.24s}.rise.d4{animation-delay:.32s}
@keyframes rise{from{transform:translateY(10px);opacity:.001}to{transform:none;opacity:1}}

/* permission sheet */
.scrim{position:fixed;inset:0;background:rgba(0,0,0,.6);opacity:0;pointer-events:none;transition:opacity
```

```
.25s;z-index:8}
.scrim.show{opacity:1;pointer-events:auto}
.sheet{position:fixed;left:0;right:0;bottom:0;z-index:9;background:var(--surface);border-top:1px solid
var(--line-strong);border-radius:16px 16px 0 0;padding:18px 20px calc(22px + env(safe-area-inset-
bottom,0px));transform:translateY(105%);transition:transform .32s cubic-
bezier(.2,.7,.2,1);display:flex;flex-direction:column;gap:12px}
.sheet.show{transform:none}
.sheet .h{font-family:var(--display);font-size:26px;line-height:26px;font-weight:700;text-
transform:uppercase}
.sheet p{margin:0;font-size:14px;line-height:20px;color:var(--fg-2)}
.sheet .row{display:flex;gap:8px;margin-top:4px}
.btn{flex:1 1 0;height:48px;border-radius:4px;border:1px solid var(--line-
strong);background:transparent;color:var(--fg);font:700 19px/1 var(--display);text-
transform:uppercase;cursor:pointer}
.btn.primary{background:var(--accent);border-color:var(--accent);color:var(--accent-ink)}

/* step 2 placeholder */
.listen{flex:1 1 auto;display:flex;flex-direction:column;align-items:center;justify-
content:center;gap:18px;text-align:center}
.mic{width:96px;height:96px;border-radius:50%;background:var(--accent);display:flex;align-
items:center;justify-content:center;box-shadow:0 0 0 14px var(--accent-soft);animation:breathe 1.6s ease-
in-out infinite}
.mic.typing{background:var(--surface);border:1px solid var(--line-strong);box-shadow:none;animation:none}
@keyframes breathe{0%,100%{box-shadow:0 0 0 12px var(--accent-soft)}50%{box-shadow:0 0 0 24px
rgba(255,110,1,.06)}}
.listen .big{font-family:var(--display);font-size:30px;line-height:30px;font-weight:700;text-
transform:uppercase}
.listen .sub{font-size:14px;line-height:20px;color:var(--fg-3);max-width:280px}
.back{align-self:center;background:none;border:0;color:var(--fg-3);font:500 13px/1 var(--body);letter-
spacing:.04em;cursor:pointer;padding:12px}
.tag{display:inline-block;font-size:10px;letter-spacing:.12em;text-transform:uppercase;color:var(--
accent);border:1px solid var(--accent);border-radius:2px;padding:3px 7px}

/* demo controls, outside the phone */
.demo{display:flex;flex-wrap:wrap;gap:8px;align-items:center;justify-content:center;padding:10px
14px;background:#111;border-top:1px solid var(--line);font-size:12px;color:var(--fg-3);max-width:100%;box-
sizing:border-box}
.demo b{color:var(--fg-2);font-weight:600;letter-spacing:.08em;text-transform:uppercase;font-size:11px}
.demo button{font:600 12px/1 var(--body);padding:8px 10px;border-radius:3px;border:1px solid var(--line-
strong);background:transparent;color:var(--fg-2);cursor:pointer}
.demo button[aria-pressed="true"]{border-color:var(--accent);color:var(--accent)}
@media (prefers-reduced-motion:reduce)
{.rise,.dot,.mic{animation:none!important}.sheet,.scrim{transition:none}}
</style>

<div class="stage">
<div class="phone" id="phone">

  <section class="screen" id="s1">
    <div class="top rise">
      <span class="wordmark">TAKOMO</span>
      <div class="units" role="group" aria-label="Units">
        <button type="button" id="u-imp" aria-pressed="true">yd · mph</button>
        <button type="button" id="u-met" aria-pressed="false">m · km/h</button>
      </div>
    </div>
    <div class="rule rise"></div>

    <div class="cap idle rise d1" id="cap">
      <div class="cap-label"><span class="dot"></span><span id="cap-label">Fitter</span></div>
      <div class="cap-text" id="cap-text" aria-live="polite"></div>
      <button type="button" class="hear" id="hear" hidden>
        <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round"
aria-hidden="true"><path d="M4 10v4h4l5 4V6L8 10H4zM16 9a4 4 0 0 1 0 6M18.5 6.5a8 8 0 0 1 0 11"/></svg>
        Tap to hear him
      </button>
    </div>

    <div class="doors">
      <h1 class="rise d2" id="h1">Find<br>your fit.</h1>
      <button type="button" class="door primary rise d2" id="d-talk">
        <span><span class="t" id="d-talk-t">Talk to the fitter</span><span class="s" id="d-talk-s">~10 min
```

```
· he measures with you and watches you swing</span></span>
        <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2.2" stroke-
linecap="round" aria-hidden="true"><rect x="9" y="3" width="6" height="11" rx="3"/><path d="M5 11a7 7 0 0
0 14 0M12 18v3"/></svg>
      </button>
      <button type="button" class="door rise d3" id="d-type">
        <span><span class="t">Same thing, typing</span><span class="s">on a train, in a meeting,
whatever</span></span>
        <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round"
aria-hidden="true"><rect x="3" y="6" width="18" height="12" rx="2"/><path d="M7 10h.01M11 10h.01M15
10h.01M8 14h8"/></svg>
      </button>
      <button type="button" class="door rise d4" id="d-quiz">
        <span><span class="t">Quick quiz</span><span class="s">~3 min · eight questions, each says why
it’s asked</span></span>
        <span class="arrow">→</span>
      </button>
      <p class="fine rise d4">No email to see your fit · it’s saved so you can come back · same engine
either way</p>
    </div>

    <div class="scrim" id="scrim"></div>
    <div class="sheet" id="sheet" role="dialog" aria-labelledby="sheet-h" aria-modal="true">
      <span class="tag">Microphone</span>
      <div class="h" id="sheet-h">He needs to hear you.</div>
      <p>Your voice is turned into text on the spot. Nothing is recorded or kept. You can switch to typing
any time.</p>
      <div class="row">
        <button type="button" class="btn" id="sheet-no">Not now</button>
        <button type="button" class="btn primary" id="sheet-yes">Allow</button>
      </div>
    </div>
  </section>

  <section class="screen" id="s2" hidden>
    <div class="top">
      <span class="wordmark">TAKOMO</span>
      <span style="font-size:11px;letter-spacing:.1em;text-transform:uppercase;color:var(--fg-3)">AI
fitter · irons</span>
    </div>
    <div class="rule"></div>
    <div class="cap" id="cap2">
      <div class="cap-label"><span class="dot"></span><span>Fitter</span></div>
      <div class="cap-text" id="cap2-text"></div>
    </div>
    <div class="listen">
      <div class="mic" id="mic">
        <svg viewBox="0 0 24 24" width="40" height="40" fill="none" stroke="#0B0B0B" stroke-width="2.2"
stroke-linecap="round" aria-hidden="true"><rect x="9" y="3" width="6" height="11" rx="3"/><path d="M5 11a7
7 0 0 0 14 0M12 18v3"/></svg>
      </div>
      <div class="big" id="s2-big">Listening</div>
      <div class="sub" id="s2-sub">Step 2 starts here. This prototype stops at the door — the chat is the
next build.</div>
      <span class="tag">Step 2 · next</span>
    </div>
    <button type="button" class="back" id="back">← Back to the start</button>
  </section>

</div>

<div class="demo" id="demo">
  <b>Demo</b>
  <button type="button" id="dm-reset">Replay</button>
  <button type="button" id="dm-return" aria-pressed="false">Returning golfer</button>
  <button type="button" id="dm-mute" aria-pressed="false">Mute voice</button>
  <span id="dm-note"></span>
</div>
</div>

<script>
(function(){
```

```
  const $ = id => document.getElementById(id);
  const cap = $('cap'), capText = $('cap-text'), hear = $('hear');
  const s1 = $('s1'), s2 = $('s2');
  const reduce = matchMedia('(prefers-reduced-motion: reduce)').matches;

  const LINES = {
    fresh: "Hey. I’m Takomo’s AI fitter — not a person, but I know these irons better than most people do. Ten minutes, I explain every spec, and I tell you when I’m guessing. Talk or type?",
    back:  "You’re back. Last time we left 201 versus 201T open. Got sim numbers, or want to pick up where we were?",
    talk:  "Right. Tell me about your game — how long you’ve played, how often, and what you’d like these irons to fix.",
    type:  "No problem, type away. Same questions, same fit. Tell me about your game.",
    quiz:  "Eight questions. I’ll say why I’m asking each one.",
    cont:  "Good. Where were we — you were going to hit both at a demo bay. Did you?"
  };

  let state = { returning:false, muted:false, audioOK:false, speaking:false, typer:null, utter:null };

  // ---- caption: types out, or highlights words in sync with the voice
  function setWords(text){
    capText.innerHTML = '';
    text.split(' ').forEach((w,i)=>{
      const s = document.createElement('span'); s.className='w todo'; s.textContent = w + ' ';
s.dataset.i=i; capText.appendChild(s);
    });
    return capText.querySelectorAll('.w');
  }
  function stopCaption(){
    if(state.typer){ clearInterval(state.typer); state.typer=null; }
    if(window.speechSynthesis){ speechSynthesis.cancel(); }
    state.speaking=false;
    capText.querySelectorAll('.w').forEach(w=>{ w.classList.remove('on','todo'); });
    cap.classList.remove('idle'); cap.classList.add('done');
  }
  function typeLine(text, done){
    const words = setWords(text);
    let i=0; cap.classList.add('idle'); cap.classList.remove('done');
    const step = reduce ? 0 : 150;
    if(!step){ words.forEach(w=>w.classList.remove('todo')); cap.classList.remove('idle');
cap.classList.add('done'); done&&done(); return; }
    state.typer = setInterval(()=>{
      if(i>0) words[i-1].classList.remove('on');
      if(i<words.length){ words[i].classList.remove('todo'); words[i].classList.add('on'); i++; }
      else { clearInterval(state.typer); state.typer=null; cap.classList.remove('idle');
cap.classList.add('done'); done&&done(); }
    }, step);
  }
  function pickVoice(){
    const vs = speechSynthesis.getVoices();
    const pref = ['Daniel','Google UK English Male','Microsoft Ryan','Alex','Google US
English','Samantha'];
    for(const p of pref){ const v = vs.find(v=>v.name.includes(p)); if(v) return v; }
    return vs.find(v=>/^en/i.test(v.lang)) || vs[0] || null;
  }
  function speakLine(text, done){
    if(!window.speechSynthesis || state.muted || !state.audioOK){ typeLine(text, done); return; }
    speechSynthesis.cancel();
    const words = setWords(text);
    cap.classList.add('idle'); cap.classList.remove('done');
    const u = new SpeechSynthesisUtterance(text);
    const v = pickVoice(); if(v) u.voice = v;
    u.rate = 1.02; u.pitch = 0.92;
    let idx = 0; const starts=[]; let pos=0;
    text.split(' ').forEach(w=>{ starts.push(pos); pos += w.length+1; });
    u.onboundary = e => {
      if(e.name && e.name!=='word') return;
      while(idx<starts.length && starts[idx] <= e.charIndex){ idx++; }
      words.forEach((w,i)=>{ w.classList.toggle('on', i===idx-1); if(i<idx) w.classList.remove('todo');
});
    };
    u.onend = () => { if(state.utter!==u) return; words.forEach(w=>w.classList.remove('on','todo'));
```

```
cap.classList.remove('idle'); cap.classList.add('done'); state.speaking=false; done&&done(); };
    u.onerror = () => { if(state.utter!==u) return; stopCaption(); typeLine(text, done); };
    state.utter = u; state.speaking = true;
    speechSynthesis.speak(u);
    // Safari sometimes never fires boundary events: fall back to a timed reveal.
    setTimeout(()=>{ if(state.speaking && !words[0].classList.contains('on') &&
words[0].classList.contains('todo')){ let i=0; state.typer=setInterval(()=>{ if(i<words.length){
words[i].classList.remove('todo'); i++; } else { clearInterval(state.typer); state.typer=null; } }, 150);
} }, 700);
  }
  function say(text, done){
    stopCaption();
    if(state.audioOK && !state.muted && window.speechSynthesis){ hear.hidden = true; speakLine(text,
done); }
    else { hear.hidden = state.muted || !window.speechSynthesis; typeLine(text, done); }
  }

  // ---- screen 1 content by state
  function renderS1(){
    const r = state.returning;
    $('h1').innerHTML = r ? 'Welcome<br>back.' : 'Find<br>your fit.';
    $('d-talk-t').textContent = r ? 'Continue where we left off' : 'Talk to the fitter';
    $('d-talk-s').textContent = r ? '201 vs 201T still open · 2 min to settle it' : '~10 min · he measures
with you and watches you swing';
    $('dm-note').textContent = r ? 'saved fit found (simulated)' : '';
  }
  function start(){
    stopCaption();
    s2.hidden = true; s1.hidden = false;
    hideSheet();
    renderS1();
    // re-run the entrance
    s1.querySelectorAll('.rise').forEach(el=>{ el.style.animation='none'; void el.offsetWidth;
el.style.animation=''; });
    setTimeout(()=> say(state.returning ? LINES.back : LINES.fresh), reduce ? 0 : 500);
  }

  // ---- doors
  function go(kind){
    stopCaption(); // barge-in: he stops mid-sentence
    if(kind==='talk'){ showSheet(); return; }
    enterS2(kind);
  }
  function enterS2(kind){
    hideSheet();
    s1.hidden = true; s2.hidden = false;
    const mic = $('mic');
    const voice = kind==='talk';
    mic.classList.toggle('typing', !voice);
    $('s2-big').textContent = voice ? 'Listening' : (kind==='quiz' ? 'Question 1 of 8' : 'Your turn —
type');
    $('s2-sub').textContent = voice ? 'Hold the button and talk. Everything he hears shows up as text so
you can correct him.' : (kind==='quiz' ? 'Same engine as the chat. He still explains every question.' :
'Same fitter, keyboard instead of mic. He still talks — you read.');
    const line = state.returning && voice ? LINES.cont : LINES[kind];
    const words = $('cap2-text'); words.textContent = line;
    if(state.audioOK && !state.muted && window.speechSynthesis){ const u=new
SpeechSynthesisUtterance(line); const v=pickVoice(); if(v)u.voice=v; u.rate=1.02; u.pitch=0.92;
speechSynthesis.speak(u); }
  }
  function showSheet(){ $('scrim').classList.add('show'); $('sheet').classList.add('show'); $('sheet-
yes').focus(); }
  function hideSheet(){ $('scrim').classList.remove('show'); $('sheet').classList.remove('show'); }

  $('d-talk').addEventListener('click', ()=>go('talk'));
  $('d-type').addEventListener('click', ()=>go('type'));
  $('d-quiz').addEventListener('click', ()=>go('quiz'));
  $('sheet-yes').addEventListener('click', ()=>enterS2('talk'));
  $('sheet-no').addEventListener('click', ()=>enterS2('type'));   // denied → typing, no second ask
  $('scrim').addEventListener('click', hideSheet);
  $('back').addEventListener('click', start);
```

```
  // first tap anywhere unlocks audio for the rest of the session (browsers need a gesture)
  hear.addEventListener('click', ()=>{ state.audioOK = true; say(state.returning ? LINES.back :
LINES.fresh); });
  document.addEventListener('pointerdown', ()=>{ if(!state.audioOK && !state.muted &&
window.speechSynthesis){ state.audioOK = true; } }, {capture:true});

  // units
  $('u-imp').addEventListener('click', ()=>setUnits('imp'));
  $('u-met').addEventListener('click', ()=>setUnits('met'));
  function setUnits(u){
    $('u-imp').setAttribute('aria-pressed', u==='imp'); $('u-met').setAttribute('aria-pressed',
u==='met');
    try{ localStorage.setItem('fitter.units', u); }catch(e){}
  }
  try{ const u = localStorage.getItem('fitter.units'); if(u) setUnits(u); }catch(e){}

  // demo controls
  $('dm-reset').addEventListener('click', start);
  $('dm-return').addEventListener('click', e=>{ state.returning = !state.returning;
e.currentTarget.setAttribute('aria-pressed', state.returning); start(); });
  $('dm-mute').addEventListener('click', e=>{ state.muted = !state.muted;
e.currentTarget.setAttribute('aria-pressed', state.muted); if(state.muted) stopCaption(); hear.hidden =
state.muted || cap.classList.contains('done'); });

  if(window.speechSynthesis){ speechSynthesis.onvoiceschanged = ()=>{}; speechSynthesis.getVoices(); }
  start();
})();
</script>
```